



K.R. Mangalam University
School of Engineering & Technology

Fundamentals Of Java Programming Lab
(ENCA203) Theory Assignment 2

Calculator Application Using Method
Overloading

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Course: BCA (Ai & Ds) Sec-B

Calculator Application Using Method Overloading:

Code:

```
calculator_assignment.java X
calculator_assignment.java > calculator_assignment
1  import java.util.Scanner;
2
3  class Calculator {
4
5      // Addition with two integers
6  public int add(int a, int b) {
7      return a + b;
8  }
9
10     // Addition with two doubles
11 public double add(double a, double b) {
12     return a + b;
13 }
14
15     // Addition with three integers
16 public int add(int a, int b, int c) {
17     return a + b + c;
18 }
19
20     // Subtraction
21 public int subtract(int a, int b) {
22     return a - b;
23 }
24
25     // Multiplication
26 public double multiply(double a, double b) {
27     return a * b;
28 }
29
30     // Division
31 public double divide(int a, int b) {
32     try {
33         if (b == 0) {
34             throw new ArithmeticException("Division by zero is not allowed!");
35         }
36         return (double) a / b;
37     } catch (ArithmeticException e) {
38         System.out.println("Error: " + e.getMessage());
39         return Double.NaN; // Not a Number (invalid result)
40     }
41 }
```

```
39     return Double.NaN; // Not a Number (invalid result)
40 }
41 }
42 }
43
44 public class calculator_assignment {
45
46     Scanner sc = new Scanner(System.in);
47     Calculator calc = new Calculator();
48
49     public void performAddition() {
50         System.out.println("\nChoose addition type:");
51         System.out.println("1. Add two integers");
52         System.out.println("2. Add two doubles");
53         System.out.println("3. Add three integers");
54         System.out.print("Enter your choice: ");
55         int ch = sc.nextInt();
56
57         switch (ch) {
58             case 1:
59                 System.out.print("Enter first integer: ");
60                 int a1 = sc.nextInt();
61                 System.out.print("Enter second integer: ");
62                 int b1 = sc.nextInt();
63                 System.out.println("Result: " + calc.add(a1, b1));
64                 break;
65             case 2:
66                 System.out.print("Enter first double: ");
67                 double a2 = sc.nextDouble();
68                 System.out.print("Enter second double: ");
69                 double b2 = sc.nextDouble();
70                 System.out.println("Result: " + calc.add(a2, b2));
71                 break;
72             case 3:
73                 System.out.print("Enter first integer: ");
74                 int a3 = sc.nextInt();
75                 System.out.print("Enter second integer: ");
```

```
76         int b3 = sc.nextInt();
77         System.out.print("Enter third integer: ");
78         int c3 = sc.nextInt();
79         System.out.println("Result: " + calc.add(a3, b3, c3));
80         break;
81     default:
82         System.out.println("Invalid choice!");
83     }
84 }
85
86 public void performSubtraction() {
87     System.out.print("\nEnter first integer: ");
88     int a = sc.nextInt();
89     System.out.print("Enter second integer: ");
90     int b = sc.nextInt();
91     System.out.println("Result: " + calc.subtract(a, b));
92 }
93
94 public void performMultiplication() {
95     System.out.print("\nEnter first double: ");
96     double a = sc.nextDouble();
97     System.out.print("Enter second double: ");
98     double b = sc.nextDouble();
99     System.out.println("Result: " + calc.multiply(a, b));
100 }
101
102 public void performDivision() {
103     System.out.print("\nEnter numerator (integer): ");
104     int a = sc.nextInt();
105     System.out.print("Enter denominator (integer): ");
106     int b = sc.nextInt();
107     double result = calc.divide(a, b);
108     if (!Double.isNaN(result)) {
109         System.out.println("Result: " + result);
110     }
111 }
```

```

112
113  public void mainMenu() {
114      int choice;
115      do {
116          System.out.println("\n--- Calculator Application ---");
117          System.out.println("1. Add Numbers");
118          System.out.println("2. Subtract Numbers");
119          System.out.println("3. Multiply Numbers");
120          System.out.println("4. Divide Numbers");
121          System.out.println("5. Exit");
122          System.out.print("Enter your choice: ");
123          choice = sc.nextInt();
124
125          switch (choice) {
126              case 1:
127                  performAddition();
128                  break;
129              case 2:
130                  performSubtraction();
131                  break;
132              case 3:
133                  performMultiplication();
134                  break;
135              case 4:
136                  performDivision();
137                  break;
138              case 5:
139                  System.out.println("Thank you for using the calculator!");
140                  break;
141              default:
142                  System.out.println("Invalid choice! Please try again.");
143          }
144      } while (choice != 5);
145  }
146
147      Run | Debug
148      public static void main(String[] args) {
149          calculator_assignment ui = new calculator_assignment();
150          ui.mainMenu();
151      }
152  }
153

```

```
import java.util.Scanner;
```

```
class Calculator {
```

```
    // Addition with two integers
```

```
    public int add(int a, int b) {
```

```
        return a + b;
```

```
    }
```

```
    // Addition with two doubles
```

```
    public double add(double a, double b) {
```

```
        return a + b;
```

```
    }
```

```
    // Addition with three integers
```

```
    public int add(int a, int b, int c) {
```

```
        return a + b + c;
```

```
    }
```

```
    // Subtraction
```

```
    public int subtract(int a, int b) {
```

```
        return a - b;
```

```
    }
```

```
    // Multiplication
```

```
    public double multiply(double a, double b) {
```

```
        return a * b;
```

```
    }
```

```
// Division
public double divide(int a, int b) {
    try {
        if (b == 0) {
            throw new ArithmeticException("Division by zero is not allowed!");
        }
        return (double) a / b;
    } catch (ArithmeticException e) {
        System.out.println("Error: " + e.getMessage());
        return Double.NaN; // Not a Number (invalid result)
    }
}
}
```

```
public class calculator_assignment {

    Scanner sc = new Scanner(System.in);
    Calculator calc = new Calculator();

    public void performAddition() {
        System.out.println("\nChoose addition type:");
        System.out.println("1. Add two integers");
        System.out.println("2. Add two doubles");
        System.out.println("3. Add three integers");
        System.out.print("Enter your choice: ");
        int ch = sc.nextInt();

        switch (ch) {
            case 1:
```

```

        System.out.print("Enter first integer: ");
        int a1 = sc.nextInt();
        System.out.print("Enter second integer: ");
        int b1 = sc.nextInt();
        System.out.println("Result: " + calc.add(a1, b1));
        break;
    case 2:
        System.out.print("Enter first double: ");
        double a2 = sc.nextDouble();
        System.out.print("Enter second double: ");
        double b2 = sc.nextDouble();
        System.out.println("Result: " + calc.add(a2, b2));
        break;
    case 3:
        System.out.print("Enter first integer: ");
        int a3 = sc.nextInt();
        System.out.print("Enter second integer: ");
        int b3 = sc.nextInt();
        System.out.print("Enter third integer: ");
        int c3 = sc.nextInt();
        System.out.println("Result: " + calc.add(a3, b3, c3));
        break;
    default:
        System.out.println("Invalid choice!");
    }
}

public void performSubtraction() {
    System.out.print("\nEnter first integer: ");

```

```
int a = sc.nextInt();  
System.out.print("Enter second integer: ");  
int b = sc.nextInt();  
System.out.println("Result: " + calc.subtract(a, b));  
}
```

```
public void performMultiplication() {  
    System.out.print("\nEnter first double: ");  
    double a = sc.nextDouble();  
    System.out.print("Enter second double: ");  
    double b = sc.nextDouble();  
    System.out.println("Result: " + calc.multiply(a, b));  
}
```

```
public void performDivision() {  
    System.out.print("\nEnter numerator (integer): ");  
    int a = sc.nextInt();  
    System.out.print("Enter denominator (integer): ");  
    int b = sc.nextInt();  
    double result = calc.divide(a, b);  
    if (!Double.isNaN(result)) {  
        System.out.println("Result: " + result);  
    }  
}
```

```
public void mainMenu() {  
    int choice;  
    do {  
        System.out.println("\n--- Calculator Application ---");
```

```
System.out.println("1. Add Numbers");
System.out.println("2. Subtract Numbers");
System.out.println("3. Multiply Numbers");
System.out.println("4. Divide Numbers");
System.out.println("5. Exit");
System.out.print("Enter your choice: ");
choice = sc.nextInt();
```

```
switch (choice) {
    case 1:
        performAddition();
        break;
    case 2:
        performSubtraction();
        break;
    case 3:
        performMultiplication();
        break;
    case 4:
        performDivision();
        break;
    case 5:
        System.out.println("Thank you for using the calculator!");
        break;
    default:
        System.out.println("Invalid choice! Please try again.");
}
```

```
} while (choice != 5);
```

```
}
```

```
public static void main(String[] args) {
```

```
    calculator_assignment ui = new calculator_assignment();
```

```
    ui.mainMenu();
```

```
}
```

```
}
```

Output:

```
--- Calculator Application ---
1. Add Numbers
2. Subtract Numbers
3. Multiply Numbers
4. Divide Numbers
5. Exit
Enter your choice: 1
```

```
Choose addition type:
1. Add two integers
2. Add two doubles
3. Add three integers
Enter your choice: 1
Enter first integer: 2
Enter second integer: 2
Result: 4
```

```
--- Calculator Application ---
1. Add Numbers
2. Subtract Numbers
3. Multiply Numbers
4. Divide Numbers
5. Exit
Enter your choice: 1
```

```
Choose addition type:
1. Add two integers
2. Add two doubles
3. Add three integers
Enter your choice: 2
Enter first double: 2.0
Enter second double: 2.0
Result: 4.0
```

```
--- Calculator Application ---
1. Add Numbers
2. Subtract Numbers
3. Multiply Numbers
4. Divide Numbers
5. Exit
Enter your choice: 1
```

```
Choose addition type:
1. Add two integers
2. Add two doubles
3. Add three integers
Enter your choice: 3
Enter first integer: 2
Enter second integer: 2
Enter third integer: 2
Result: 6
```

--- Calculator Application ---

1. Add Numbers
2. Subtract Numbers
3. Multiply Numbers
4. Divide Numbers
5. Exit

Enter your choice: 2

Enter first integer: 2

Enter second integer: 2

Result: 0

--- Calculator Application ---

1. Add Numbers
2. Subtract Numbers
3. Multiply Numbers
4. Divide Numbers
5. Exit

Enter your choice: 3

Enter first double: 2

Enter second double: 2

Result: 4.0

--- Calculator Application ---

1. Add Numbers
2. Subtract Numbers
3. Multiply Numbers
4. Divide Numbers
5. Exit

Enter your choice: 4

Enter numerator (integer): 4

Enter denominator (integer): 2

Result: 2.0

--- Calculator Application ---

1. Add Numbers
2. Subtract Numbers
3. Multiply Numbers
4. Divide Numbers
5. Exit

Enter your choice: 5

Thank you for using the calculator!

Explanation:

1. Concept Used: Method Overloading

- **Definition:** When multiple methods in the same class have the same name but different parameter lists (number, type, or order), it's called **method overloading**.
- It's a form of **compile-time polymorphism** in Java.

2. Structure of the Program

- **Class 1 → Calculator:**
 - Contains all arithmetic operations as overloaded methods.
 - Demonstrates compile-time polymorphism.
 - Includes exception handling for division (avoids crash on divide-by-zero).
- **Class 2 → UserInterface:**
 - Takes user input using a Scanner object.
 - Displays a **menu-driven system** for easy interaction.
 - Calls methods from the Calculator class.
 - Handles multiple data types (int, double) based on user choice.

3. Exception Handling

Used in:

```
public double divide(int a, int b)
```

If the denominator is zero, an Arithmetic Exception is thrown and caught with a message "Division by zero is not allowed!".

4. Control Structures

- **Switch statements** for operation selection.
- **Do-while loop** for menu repetition until the user chooses to exit.
- Ensures the program runs continuously until choice == 5.